

MATH 3311, FALL 2026: HOMEWORK 1

Much of the material in this homework is very classical and all over the interweb. So I urge you to resist the call of Google or ChatGPT, and work everything out for yourself. It'll be *much* more rewarding that way. Even if this resistance proves ultimately futile, keep in mind that the exams are in-class with no assistance, LLM-flavored or otherwise, so you should at least understand what you're writing down.

Now, to business: Long division (that is, the Euclidean algorithm) says that if $a, b \in \mathbb{Z}$, with $b > 0$, then we can find $q, r \in \mathbb{Z}$ with $|r| < b$ such that

$$a = qb + r.$$

Consider the following process: Set $r_0 = r$, and apply the Euclidean algorithm to the pair b, r_0 to get q_1, r_1 such that

$$b = q_1 r_0 + r_1.$$

Then apply the Euclidean algorithm again to the pair r_0, r_1 :

$$r_0 = q_2 r_1 + r_2,$$

and then to the pair r_1, r_2 , and so on. At each stage, if you prefer, you can take r_i to be the *positive* remainder.

- (1) Show that the process terminates; that is, for some natural number $n \geq 1$, we must get $r_n = 0$. Show also that $r_{n-1} = \gcd(a, b)$, where $\gcd(a, b)$ denotes the unique nonnegative common divisor of a and b having maximal absolute value.

- (2) Use the above process to conclude that we can find integers $s, t \in \mathbb{Z}$ such that

$$\gcd(a, b) = sa + tb.$$

- (3) If $b = p$ is a prime number and $p \nmid a$, show that we can find integers $s, t \in \mathbb{Z}$ such that

$$1 = sa + tp.$$

An *equivalence relation* on a set X is a relation $x \sim y$ on pairs of elements $x, y \in X$ with the following properties:

Reflexivity: $x \sim x$ for all $x \in X$;

Symmetry: $x \sim y \Leftrightarrow y \sim x$;

Transitivity: $x \sim y$ and $y \sim z \Rightarrow x \sim z$.

If $\sim$ is an equivalence relation on X and $x \in X$, the *equivalence class* of x is the subset

$$[x] = \{y \in X : y \sim x\} \subset X.$$

An element x of an equivalence class is called a *representative* for that class. A *full set of representatives* is a subset $P \subset X$ containing exactly one representative from each equivalence class. Thus P is in bijection with the set of equivalence classes and can, with some care, be used as a stand-in for that set.

- (4) Let n be a positive integer. Show that the relation

$$x \equiv y \pmod{n}$$

defined by the condition $n \mid x - y$ is an equivalence relation on $\mathbb{Z}$, and show that the subset

$$\{0, 1, 2, \dots, n-1\} \subset \mathbb{Z}$$

is a full set of representatives for this relation.

- (5) Let $\mathbb{Z}/n\mathbb{Z}$ be the set of equivalence classes for the relation $x \equiv y \pmod{n}$. It is in bijection with the set $\{0, 1, 2, \dots, n-1\}$ by the previous problem.

We would like to define addition and multiplication by

$$[a] + [b] = [a + b] \quad \text{and} \quad [a] \cdot [b] = [ab].$$

First show that these definitions do not depend on the choice of representatives: If

$$a \equiv a' \pmod{n} \quad \text{and} \quad b \equiv b' \pmod{n},$$

show that

$$a + b \equiv a' + b' \pmod{n} \quad \text{and} \quad ab \equiv a'b' \pmod{n}.$$

Thus addition and multiplication are well-defined operations on $\mathbb{Z}/n\mathbb{Z}$.

For instance, in $\mathbb{Z}/n\mathbb{Z}$, the product $2 \cdot (n-1)$ is represented by $n-2$, while the sum $2 + (n-1)$ is represented by 1.

Now suppose that $n = p$ is prime. Show that, for any $a \in \mathbb{Z}/p\mathbb{Z}$ with $a \neq 0$, the equation

$$ax = 1$$

admits a solution $x \in \mathbb{Z}/p\mathbb{Z}$.

- (6) Let p be prime and let $a \in \mathbb{Z}/p\mathbb{Z}$ be nonzero. Show that

$$\{a, 2a, 3a, \dots, (p-1)a\} = \{1, 2, \dots, p-1\}$$

as subsets of $\mathbb{Z}/p\mathbb{Z}$.

- (7) Conclude that

$$a^{p-1} \cdot (p-1)! = (p-1)!$$

in $\mathbb{Z}/p\mathbb{Z}$. Explain why $(p-1)! \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$, and use the previous problems to justify cancelling $(p-1)!$ from both sides. Deduce Fermat's little theorem: For any integer $a \in \mathbb{Z}$ with $p \nmid a$,

$$p \mid a^{p-1} - 1.$$

- (8) Suppose that you have a deck of n cards, where n is even. A *perfect riffle shuffle* (see: <https://www.youtube.com/watch?v=adNeBLkP8ZM>) is obtained by cutting the deck into two equal halves and then interleaving the two halves in alternating fashion. There are two ways to do this, depending on which half contributes the first card: one leaves the original top card on top, while the other sends it to the second position.

Can every permutation of the n cards be realized by applying a sequence of perfect riffle shuffles of these two types? Experiment with some small values of n , formulate a conjecture, and prove your answer in at least one nontrivial case.

The next problem requires the notions of *group* and *group homomorphism*, which will be introduced in class on Wednesday.

First, a definition: For any set X , write $\text{Fun}(X)$ for the set of functions $f : X \rightarrow X$ from X to itself. Within it, we have the subset $\text{Bij}(X)$ of functions f that are *bijective*.

(9) Show that function composition defines an operation

$$\circ : \text{Fun}(X) \times \text{Fun}(X) \longrightarrow \text{Fun}(X)$$

with the following properties:

- It is *associative*:

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

- If $\text{Id} : X \rightarrow X$ is the identity function $\text{Id}(x) = x$, then

$$\text{Id} \circ f = f \circ \text{Id} = f$$

for all $f \in \text{Fun}(X)$.

- If $f \in \text{Fun}(X)$ and there exists $\tilde{f} \in \text{Fun}(X)$ satisfying

$$f \circ \tilde{f} = \text{Id},$$

then f is surjective and $\tilde{f}$ is injective.

- Suppose that $f \in \text{Bij}(X)$. For each $y \in X$, show that there exists a unique $x \in X$ such that $f(x) = y$. Use this to define a function

$$f^{-1} : X \rightarrow X.$$

Show that

$$f \circ f^{-1} = \text{Id} \quad \text{and} \quad f^{-1} \circ f = \text{Id}.$$

Finally, show that f^{-1} is the unique function with these properties, and that $f^{-1} \in \text{Bij}(X)$. Conclude that $\text{Bij}(X)$, equipped with the composition operation, is naturally a group.